Multi-feature Musical Instrument Sound Classifier w/user determined generalisation performance

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Abstract

A previously developed musical instrument sound classification system has been extended in the following ways: (a) three additional features have been extracted (cepstral coefficients, spectral centroid and presence of vibrato), (b) automatic instrument elimination, and (c) hierarchic and hybrid architectures introduced. Testing the system using the leave-one-out strategy with recordings taken from the same instruments used during system development has yielded a classification accuracy of 89-92% and reliability of 89-92% depending on the classification architecture used. Instrument families were identified with an accuracy of 96%, while sustain versus impulsive instruments were identified with 100% accuracy. Extensive generalisation testing using a single stage classifier over three different collections of instrument recordings showed that the system classification accuracy varied from 33% up to 61%. Applying a number of enhancements to improve the generalisation performance of the system improved the classification results by anywhere from 13 to 50 percentage points. Across four different collections of recordings, the single stage classifier consistently achieved a classification accuracy of 87-90% when the leave-one-out test strategy was used. Using instead the hierarchic classifier, a further 3% improvement would be possible. This shows that when allowed to do so, the system homes in remarkably well to the defining characteristics of any given set of test recordings. The user of the system is given direct control over its performance, which is dictated by (a) which classifier architecture is used, and (b) how much information the user is willing to provide it prior to classification being attempted.

1 Introduction

In the last three years, there has been an explosion of interest in developing a classification system that reliably identifies monophonic sounds produced by musical instruments.

Fujinaga and MacMillan (2000) designed a computer-based classifier to recognise simple continuous tones from the MUMs CDs of musical instrument sounds. 39 different timbres from 23 orchestral instruments played at different pitches were used as samples for an exemplar-based learning system that incorporated a *k*NNC with genetic algorithm.

The weighted features used during classification included moments calculated from both the steady-state and dynamically changing spectral envelopes as well as spectral irregularity and tristimulus features describing the spectral envelopes. While the recognition accuracy varied greatly between the instruments, the average overall accuracy was 68%.

Eronen (2000) developed a pitch independent isolated tone musical instrument recognition system that was tested using the full pitch range of thirty orchestral instruments from the string, brass and woodwind families, played with different articulations. Thirty-two spectral and temporal features were employed during classification. Both hierarchic and direct forms of classification were evaluated using 1498 test tones obtained from the McGill University Masters Samples (MUMs) CDs as well as from amateur musicians.

Best results were obtained with direct classification, where instrument families were classified with 95% accuracy and individual instruments with 81% accuracy.

Eronen (2001) extended his earlier work to develop a classification system that allowed him to

compare several features with regard to musical instrument recognition performance. In addition to the features described earlier, the following features were also evaluated: (i) mel-frequency cepstral coefficients, (ii) linear prediction cepstral coefficients calculated with both uniform and warped frequency scales, (iii) delta cepstral coefficients and (iv) linear prediction analysis reflection coefficients.

The features were evaluated using a database of 5286 acoustic and synthetic solo tones from 29 different orchestral instruments, out of which sixteen were used for testing purposes. Five different sources of recordings were used: (i) MUMs, (ii) acoustic guitar recordings made at Tampere University of Technology, (iii) University of Iowa, (iv) IRCAM Studio online (SOL), and (v) a Roland XP-30 synthesizer.

The best performing combination of features included mel-frequency cepstral coefficients calculated over both the onset and the steady state and a subset of the earlier spectral and temporal features. These features allowed 35% of individual instruments and 77% of instrument families to be identified correctly. The results however were found to vary across the different sources of recordings, ranging from 21% for the synthetic tones upto 87% for the University of Iowa recordings.

Agostini et al (2001) evaluated a number of classification techniques to determine which provided the lowest error rate when classifying monophonic sounds from 27 different musical instruments. The classification schemes tested were canonical discriminant analysis, quadratic discriminant analysis, support vector machines and nearest neighbours.

For each instrument sound analysed, eighteen temporal and spectral features were extracted. The test results showed that quadratic discriminant analysis provided the best results, with an accuracy of 93% for all 27 instruments. Instrument families were classified with an accuracy of 97% while pizzicato versus sustain instruments were classified correctly 97%.

In their most recent work, Kostek and Czyzewski (2001) compared the effectiveness of 23 wavelet based features derived from the attack portion of musical tones to fourteen temporal and spectral features used in previous musical instrument sound classification experiments. Experiments were carried out using a two-layer artificial neural network (ANN) having many similar characteristics to that used in their earlier work. Eighteen different instruments were tested, but taken four at a time.

In all the experiments carried out, the recognition accuracy was found to be always higher using the original fourteen features than when using the wavelet-based features. Significant variation in classification accuracy was also found to occur between instrument groups. For example, using the original fourteen features, classification accuracies ranged from 80% to 99% depending on which group of four instruments was used.

Wieczorkowska (1999a and 1999b) explored the use of both decision trees and rough sets for automatically classifying sounds from eleven orchestral (string, woodwind and brass) musical instruments played with different articulations. 62 features were used during classification. These were extracted from each sound at the beginning, middle and end of the attack as well as during the maximum and minimum amplitude portions of the steady state.

Wieczorkowska found that the rough sets outperformed the decision tree. Using a validation strategy employing 10% testing data with the rough sets achieved 90% classification accuracy.

In later work, Wieczorkowska (2000) used the same two classifiers to evaluate the use of the wavelet transform as a feature for automatically identifying musical instrument sounds. 162 features were used during classification, representing various wavelet coefficient values for different filter types taken from both the attack and steady of each sound. Out of a total of 1358 recordings, 70% were used for classifier training and 30% for testing.

In contrast to her earlier work, the experimental results showed that in general, the decision tree classifier outperformed the rough set classifier. In terms of individual instrument classification, taking into account articulation style, the rough set classifier yielded a classification accuracy of 43% compared to the decision tree classifier's 58% classification accuracy.

These results show that great progress has been made in developing a reliable musical instrument sound classifier. Some deficiencies, however, still remain. The results obtained have been generally limited to orchestral instruments. None of the studies for example has incorporated percussive instruments, nor both non-vibrato and vibrato recordings of the same instrument (cello, flute and violin). In addition,

a very limited amount of work has been carried out on the generalisation capabilities of the systems developed using more than one set of instrument recordings. Finally, no work has been done to the knowledge of the author in adapting the classification systems developed to the particular set of instrument recordings being used during testing, in an attempt to improve the generalisation performance of the system. The classification system to be described addresses these shortcomings.

2 Data Collection

Sounds from nineteen musical instruments of definite pitch were chosen for system development and testing. 604 recordings at a single dynamic level were taken directly from the MUMS CDs, of which 517 were non-vibrato and the remainder vibrato recordings. The instruments included: guitar (plucked string), violin, cello and double bass (bowed string), piano (struck string), flute (air reed wind), accordion, clarinet, saxophone (single mechanical reed wind), oboe and bassoon (double mechanical reed wind), organ (air/mechanical reed wind), trumpet, trombone, French horn, and tuba (lip reed wind) and xylophone, glockenspiel and marimba (percussive definite pitch). Both vibrato and non-vibrato recordings were used for cello, flute and violin.

The note range used was C3-C6 of the equally tempered musical scale. A three-octave note range was chosen to limit the system complexity, to provide practical limits on the number of recordings required and to simplify instrument selection. Not all instruments covered the complete note range, but each had at least some notes falling within it.

For generalisation testing, three additional sets of recordings were used. The first set of recordings were obtained from the Creative Essentials (CE) set of sound collection CDs, codeveloped by the companies Zero G and Time+Space (http://www.timespace.com/index1.asp). Recordings were provided at a single dynamic level and available for only certain notes for instruments: bassoon, cello, clarinet, flute, guitar, horn, oboe, piano, trombone, trumpet and violin. In total, 46 non-vibrato and 23 vibrato recordings were available.

The second set of recordings was obtained from Ircam, the French institute research centre into Acoustics and Music, via the Studio Online (SOL) project (http://www.ircam.fr/produits/technologies/sol/index-e.html). For consistency with the MUMs and CE collections of instrument recordings, only one dynamic level, i.e. *forte*, was used from this set. Recordings were used for the following instruments: bassoon, cello, clarinet, double bass, flute, guitar, horn, oboe, saxophone, trombone, trumpet, tuba and violin. In total, 333 non-vibrato and 61 vibrato recordings were available across the complete chromatic scale for each instrument.

The final set of recordings was kindly supplied by Prof. Kostek from the Sound and Vision Engineering Department, Faculty of Electronics, Telecommunication and Informatics, Technical University of Gdansk (TUG), Poland. The whole chromatic scale characteristics of the following instruments were used: bassoon, cello, clarinet, French horn, oboe, saxophone, trombone, trumpet and tuba. In total, 312 non-vibrato and 54 vibrato *forte* recordings were available.

3 Feature Extraction

The original classification system developed utilised only three features extracted from each musical recording (Kaminskyj 2000). These were (a) Root Mean Square (RMS) amplitude envelope, (b) Constant Q transform (CQT) frequency spectrum, and (c) Multidimensional Analysis Scaling (MSA) trajectories (see (Kaminskyj 2000) for more details). In order to improve the performance of the system, three additional features were added. These included (d) cepstral coefficients, (e) spectral centroid and (f) presence of vibrato. They were chosen either because they had been used successfully in previous research in solving similar problems (Brown 1999) or had proven important in allowing humans to discern between musical instrument sounds (Handel 1995) or simply looked intuitively promising.

3.1 Cepstral Coefficients

The cepstrum is the Fourier transform of the log magnitude spectrum of the musical sound waveform. There exist many different ways of calculating the cepstral coefficients, principally determined by the spectral measure used. The CQT spectrum with quartertone spaced spectral bins mentioned above was used to calculate the cepstral coefficients (CC_n):

$$CC_n = \sum_{k=0}^{176} \log(X[k]) \cos[n(k+0.5)\frac{\pi}{177}]$$

where: $X[k]$ is the CQT amplitude value for

spectral bin *k*, and

$n = 1..176$ since CC_{177} is always zero.

Although this equation produces 176 cepstral coefficients (ignoring the final zero coefficient), it was determined empirically that for instrument classification purposes, using only the first eleven coefficients produced the best classification results. Figure 1 shows the resultant waveform formed by the first eleven coefficients for piano playing C4.

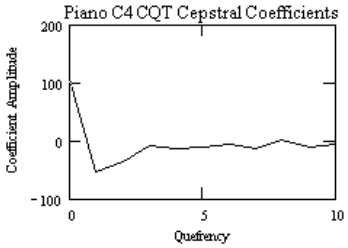


Figure 1 CQT spectrum cepstral coefficients waveforms for piano note C4

3.2 Spectral Centroid

The spectral centroid (S_k), also called brightness, is essentially a measure of the 1st moment of the spectral energy distribution. It is calculated over the duration of musical instrument recording using:

$$S_k = \frac{\sum_{j=0}^{176} A_{j,k} f_j}{\sum_{j=0}^{176} A_{j,k}}$$

where: S_k is the spectral centroid at time interval *k*,

$A_{j,k}$ is the amplitude of CQT spectral bin *j* at time interval *k*, and

f_j is the frequency of CQT spectral bin *j*

The time varying spectral centroid for four piano notes C3, C4, C5 and C6 is shown in Figure 2.

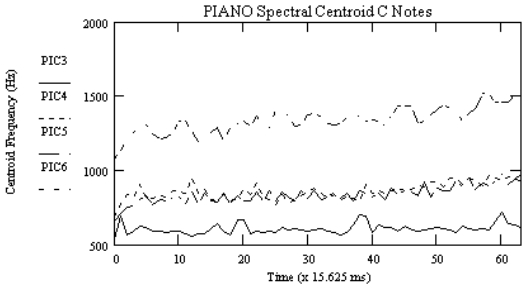


Figure 2 Time-varying spectral centroid musical for various C notes of the Piano.

3.3 Vibrato Detection

A vibrato detection system was developed to automatically detect the presence/absence of vibrato in a musical instrument recording (Kaminskyj 1998). A number of enhancements were however made in order to improve its reliability. As a result, the following conditions were required to be satisfied to qualify a recording as have been played with vibrato:

- the recording was produced by a sustain instrument,
- the peak-to-peak amplitude of the frequency variation of the largest amplitude harmonic exceeded 20 cents, and
- the amplitude fluctuation of the tremolo waveform or the frequency variation of the largest amplitude harmonic following two point moving average filtering, was in the range 5 – 8 Hz.

These conditions were based on the assumption that only cello, flute and violin were capable of being played with vibrato. If necessary, this number and range of instruments can be easily modified if required by the user.

4 Classifier Operation

The system was developed using the MUMs recordings and its structure is shown in Figure 3. The vibrato detector firstly determines the presence of any vibrato within the input test sample. If vibrato is deemed to be present, the vibrato instrument classifier is invoked, which classifies the input as having been produced by one of three possible vibrato instruments: cello, flute and violin. Alternatively, if no vibrato exists, the non-vibrato instrument classifier is executed. It classifies the input as having been produced by one of nineteen possible non-vibrato instruments.

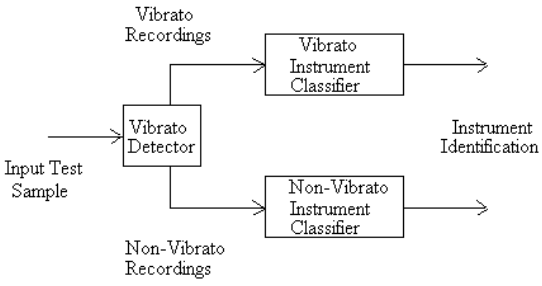


Figure 3 Classification system structure

Given the increased complexity of the non-vibrato instrument classifier, in terms of the number of

instruments it needs to classify, three different classifier architectures were evaluated: (a) single stage, (b) hierarchic and (c) hybrid. It is intended that the user of the system will be able to decide which s/he wishes to use, balancing the performance achieved with the computational effort expended.

The simplest, called the single stage classifier, in shown in Figure 4 and uses five single feature classifiers to independently determine the most likely instrument to have produced the input test sample. Each classifier employs the *k* nearest neighbour classification (*k*NNC) algorithm (Duda & Hart 1973) using one of the remaining five features listed above. The results of all five classifiers are then combined using a confusion matrix weighting scheme which makes maximum use of the known strengths and weaknesses of each classifier. The resultant combined classification results provide the best estimate of the instrument that produced the input test sample.

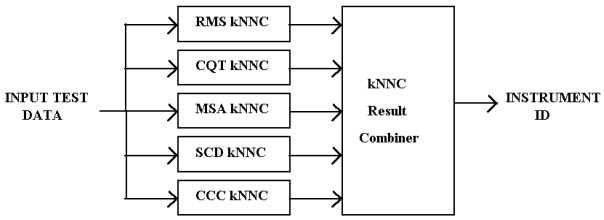


Figure 4 Single Stage Classifier

In an attempt to improve on the classification results obtained using the single stage classifier, both the hierarchic and hybrid architectures were evaluated. These used the same classifier structure as the single stage as a building block to form a taxonomic hierarchic classification tree. Each level of this tree essentially comprised a single stage classifier, but classifying a different collection of instruments. At the top of the tree of the hierarchic architecture shown in Figure 5, instruments were simply classified as being sustain or impulsive. As classification progressed down the tree, the test sample was classified into smaller and smaller groups of instruments, until finally a leaf was reached. This leaf indicated directly the most likely instrument to have produced the test input.

The hybrid classifier shown in Figure 6 comprised some of the features of both the single stage and hierarchic classifier, yielding essentially a compromise between speed of classification and performance.

Given that only three instruments were capable of producing vibrato recordings, the vibrato instrument classifier simply comprised a single stage classifier, with the same structure as described above for the non-vibrato instrument classifier.

5 Results

For the non-vibrato instrument classifier, the classification accuracy of the three classifiers evaluated was single stage 89%, hybrid 89% and hierarchic 92%. Given that the computational complexity increased dramatically when moving from the single stage classifier to the hybrid and then to the hierarchic classifier, the single stage classifier provided the best compromise in terms of classification accuracy versus computational effort. In terms of instrument family classification accuracy, both the single stage and hierarchic classifiers achieved 96%. Finally, the hierarchic classifier achieved a 100% classification accuracy for sustain versus impulsive instruments.

For the vibrato instrument classifier, 100% classification accuracy was achieved using the single stage classifier. Therefore, the overall MUMs classification accuracy taken across both the non-vibrato and vibrato recordings using the leave-one-out test strategy ranged from 90 – 93%, depending on the non-vibrato instrument classifier used.

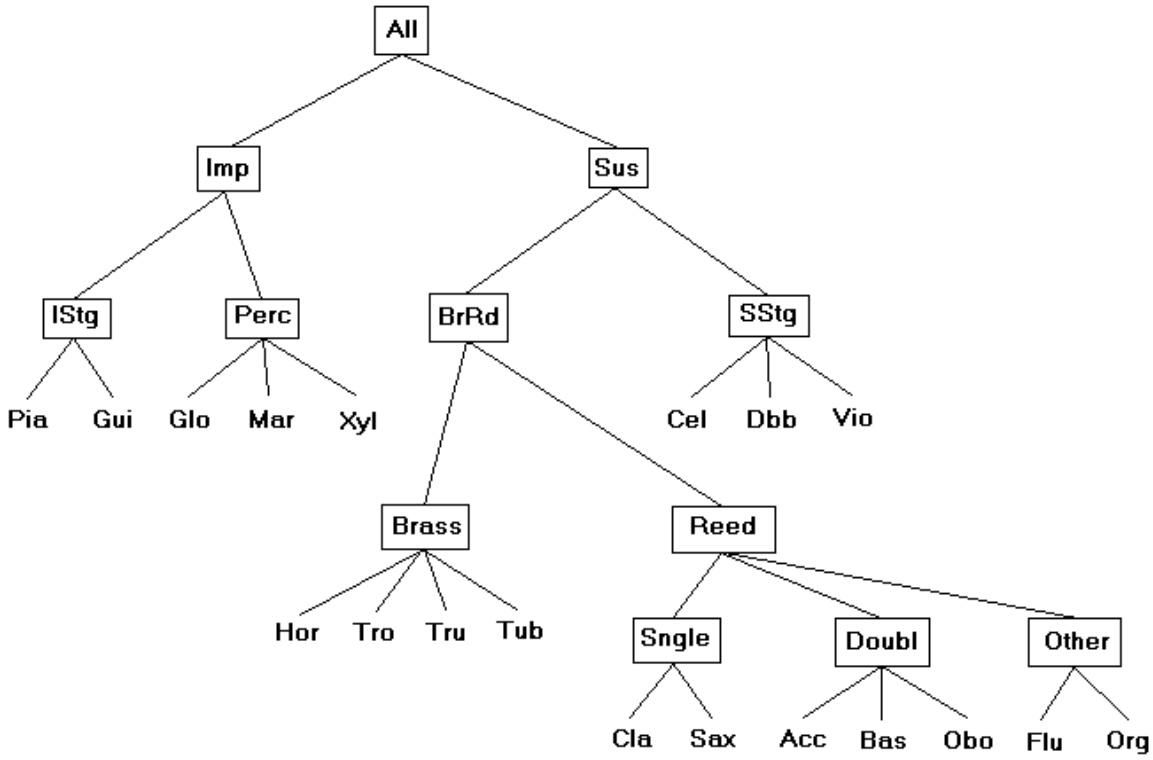


Figure 5 Hierarchic Classifier architecture

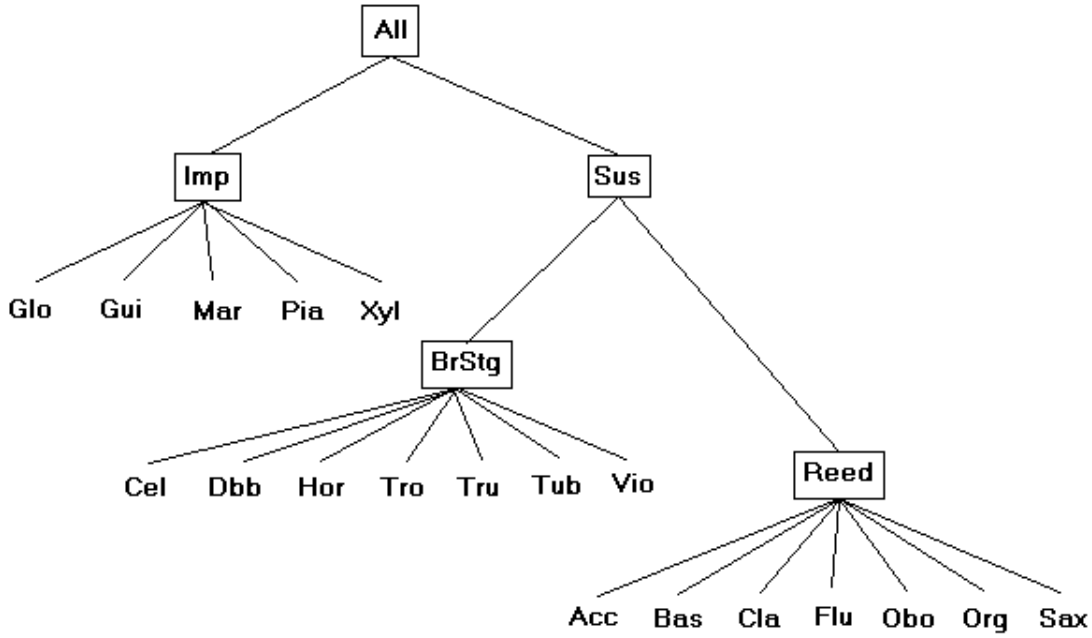


Figure 6 Hybrid Classifier architecture

The generalisation performance of the system was measured using the single stage classifier for both the vibrato and non-vibrato instrument classifiers for the three sets of recordings described above: (a) CE, (b) SOL and (c) TUG. These recording sets comprised of different combinations of vibrato and non-vibrato instrument tones. Direct classification of these sets of recordings, where the MUMs recordings were used as references by both the non-vibrato and vibrato instrument classifiers, provided a wide variety of results, ranging from 33-61%. These differences were attributed to uncontrollable factors during recording

production such as different recording environments, players, playing styles, articulations, volume levels and instrument types (e.g. acoustic vs. electric guitar).

In order to make the classification system adaptive to the defining characteristics of any individual set of recordings, and thereby improve its generalisation performance, four enhancements were explored. Firstly, any instruments known to be absent from the test recording sets were excluded from the MUMs reference set during classification. In most cases, the user of the system would know in advance what instruments s/he was working with and attempting to classify. Secondly, a small collection of test recordings, spaced at third octave intervals, was added to the MUMs reference set. In practice, the user would simply need to provide the system with a small number of labeled recordings from the instrument s/he was attempting to classify. Thirdly, the first and second enhancements were combined. As a fourth enhancement, each of the test recording sets was classified using the leave-one out test strategy, with the reference set comprising the test set recordings. Therefore, to apply this enhancement, the user would be required to supply the system with a complete collection of labeled recordings for all instruments that were being classified.

Table 1 summarises the results of applying the first three enhancements to (a) the vibrato and non-vibrato classifiers separately as well as (b) when their results were combined. Due to the small number of vibrato recordings available, the fourth enhancement was only applied to the non-vibrato recordings.

The first enhancement provided anywhere from 13 – 25 percentage point improvement in classification accuracy. For example, for the CE test set, the overall classification accuracy improved from 61% to 74% The second enhancement provided improvements in the range 36 – 39 percentage points. Given the small number of CE recordings (69), it was not feasible to apply this enhancement to these recordings. Combining these two enhancements provided further improvements in the range of 3-12 percentage points. By this stage, the classification accuracy obtained across all sets of test set recordings ranged from 74-83%. Finally, using the fourth enhancement provided classification accuracies for the non-vibrato recordings in the range 87-90%. This compared very well with the 89% classification result obtained with the non-vibrato MUMs recordings using the same leave-one-out testing strategy. As these generalisation results were obtained using the single stage classifier, a further 3% improvement could be reasonably expected when using the hierarchic classifier instead.

Table 1 Overall Generalisation performance for NVIB & VIB recordings for each recording set

Recording Set	Enhancement	Classification Performance		
		NVIB (N (%))	VIB (N (%))	Overall (N (%))
CE (Nt=69) Originally Nnv=46/Nv=23 After Vib Det Nnv=63/Nv=6	Standard	38 (60.3)	4 (66.7)	42 (60.9)
	RIE	47 (68.3)	4 (66.7)	51 (73.9)
	TstRef	N/A	N/A	N/A
	Combined	N/A	N/A	N/A
SOL (Nt=394) Originally Nnv=333/Nv=61 After Vib Det Nnv=352/Nv=42	Standard	134 (38.1)	33 (85.0)	167 (42.4)
	RIE	186 (52.8)	33 (85.0)	219 (55.6)
	TstRef (Nt=292) Nnv=261/Nv=31	203 (77.8)	27 (90.0)	230 (78.8)
	Combined (assume RIE N/A)	213 (81.6)	27 (90.0)	240 (82.2)
TUG (Nt=366) Originally Nnv=312/Nv=54 After Vib Det *Couldn't be run*	Standard	101 (32.4)	19 (35.2)	120 (32.8)
	RIE	156 (50.0)	54 (100.0)	210 (57.4)
	TstRef (Nt=320) Nnv=272/Nv=48	185 (68.0)	33 (68.8)	228 (71.3)
	Combined	211 (77.6)	54 (100.0)	265 (82.8)

Note:
1. For TUG, since only cello vibrato recordings were available, applying RIE always provided 100%

classification accuracy.

2. For TUG, due to short duration recordings available, vibrato detector could not be run.
3. For SOL combined, assumption for vibrato recordings is that since RIE provided no improvement in results on its own, then the same will occur when it is combined with the TstRef enhancement.

LEGEND

NVIB: Non-vibrato; VIB: vibrato; N: number of recordings; % : percentage correctly identified
Ntot: total number of vibrato and non-vibrato recordings; Nnv: total number of non-vibrato recordings
Nv: total number of vibrato recordings; Vib Det: vibrato detection; RIE: reference instrument elimination enhancement; TstRef: enhancement incorporating test tones at third octave spacings into the reference set

6 Discussion

The vibrato detection results summarised in the first column of Table 1 show that the vibrato detector is very good at detecting correctly non-vibrato recordings. Across 896 non-vibrato recordings from three sets of recordings (MUMs, CE and SOL), only two were incorrectly deemed to have been played with vibrato. This corresponds to an accuracy of 99.8%.

Where the vibrato detector had some difficulty related to the correct detection of recordings that were deemed to have been played with vibrato. Across 171 vibrato recordings from the same three sets of recordings, only 119 were correctly identified as having been played with vibrato. This corresponds to an accuracy of 69.6%. The overall detection accuracy across all 1067 vibrato and non-vibrato recordings was 94.9%.

The difficulty the vibrato detector had in detecting vibrato recordings was not based on any fundamental shortcoming of the detector. It brings to light instead the lack of agreement amongst musicians as to what constitutes a vibrato recording. Many of the vibrato recordings simply did not satisfy *all* the detection threshold criteria of the vibrato detector.

Many of the vibrato recordings that were incorrectly classified satisfied some of the "vibrato present" criteria and not others. These could therefore be thought of as being "weak vibrato" recordings. For example, the following SOL recordings contained frequency modulation of the correct frequency (between 5-8 Hz), were deemed to be sustain instrument recordings, but the pk-pk vibrato amplitude was well below 20 cents: cello notes A3#, E4, C3#, and violin notes F5#, G5, D5 and F4. Having the parameters of the vibrato detector user-adjustable, however seems to be most appropriate solution to cope with such a situation.

The classification results of the non-vibrato instrument classifier with the MUMs recordings show that the more effort expended during the classification process, the better the results obtained. A minor improvement in performance was however obtained for a significant increase in computational effort when moving from the single stage to the hybrid to the hierarchic classifiers. The single stage classifier provided the best compromise in terms of performance achieved with computational effort expended. These results comparing the relative performance of direct versus hierarchic classification are consistent with those obtained by Martin and Kim (1998) but contrary to those obtained by Eronen (2000).

The direct generalisation results obtained are consistent with those obtained by Eronen (2001). Eronen is the only researcher who studied the generalisation performance of his system across many sets of recordings. As mentioned in the introduction, he found that on average, only 35% of instruments were correctly identified when many sets of recordings were used. This compares well with the direct generalisation results obtained (33-61%) using a totally different classification system. This further supports the notion that most of the challenge in developing classification system is in achieving good generalisation performance, across many sets of recordings, not just the one set used during system development.

The generalisation results obtained with the system enhancements indicate, not suprisingly, that the more information the user is able and willing to provide the system with regard to the test set being used, the better the generalisation performance that is obtained. This information not only includes any knowledge of the instruments present within the test set, but also any example tones of the various instruments present. The more of these example tones

included within the reference set, the better the generalisation results become. The system has shown itself to be remarkably sensitive to the defining characteristics of the specific instruments present within the test set. This is demonstrated by the consistent 87-90% classification results obtained with the single stage classifier across four sets of recordings using the leave-one-out test strategy.

The phenomenon of classification systems being made adaptive to improve their performance concurs with the everyday experiences of speech recognition software users. These systems only begin to offer reasonable performance when trained on the peculiar characteristics of the speaker using it.

7 Conclusion

The results obtained using the classification system developed are consistent with those obtained by a variety of researchers working with a significant number of musical instruments (more than fifteen). It appears that given enough features, it is possible to achieve classification accuracies of the order of 90% or higher with a variety of classification algorithms. These results however have only been obtained using a single collection of instrument recordings for both system development and testing. The main challenge in classification system development, therefore lies in achieving good generalisation performance, across multiple collections of recording collections. This is particularly so given that musicians are most creative in finding new ways of producing different sounds from their instruments. One could argue that this is one of the reasons why music is so interesting to listen to.

In achieving this goal, two possible directions are available. The first approach would be to continue adding more features to the classification system in the hope that the new features would become increasingly less sensitive to any peculiar characteristics of any individual musical instruments. It is the belief of the author that this direction is futile. There exists too much variability in the sounds produced by musical instruments to make such a system practical.

The second approach borrows heavily from the research carried out in developing speech recognition systems. It relies instead on developing a system that is adaptive to the peculiar characteristics of the instruments being classified. The user is given the choice in defining what level of performance s/he would be happy with, based on the amount of information s/he is willing to provide the system. This is the approach currently favoured by the author.

To this end, a system architecture has been developed that works well over a significant number of general instruments. For example, it is not restricted to orchestral non-percussive instruments. Instruments that can be played both with and without vibrato can be classified whether they are played with vibrato or not. If required, the system can be easily expandable to incorporate more features during classification, simply by adjusting the number of individual feature classifiers and the classifier combiner. If necessary, it can also be easily adapted to recognise more instruments playing across a wider range of notes, volume levels, articulation types, simply by including more instruments recordings into the reference set. Most importantly, the system has scalable generalisation performance, where an improvement in results can be achieved based on the amount of effort put in by the user.

The system developed thereby represents but one step in the development of an adaptive classification system, which when properly trained, provides acceptable levels of performance.

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